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EEL 3135 – Classwork #10

Milestone:

- Complete any two questions.

Question #1: Use the DTFT definition

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

to answer the following.

- For $x[n] = 6\delta[n-2]$, derive $X(e^{j\omega})$ directly from the definition. Sketch $|X(e^{j\omega})|$ on $[-\pi, \pi]$.
- For $x[n] = a^n u[n]$ with $a = -\frac{1}{3}$, evaluate the DTFT by turning the sum into a geometric series. Clearly state any convergence condition you use.
- (Using MATLAB)** Numerically approximate the DTFT magnitude of part (b) by truncating the sum to $n = 0:40$ and sweeping $\omega \in [-\pi, \pi]$. Plot $|X(e^{j\omega})|$. Then, on the plot or in a sentence, comment on how the numerical result compares with *your* closed-form from part (b).

Question #2: Use standard DTFT pairs (e.g., for $a^n u[n]$, time shift, linearity) and do *not* rederive them from scratch.

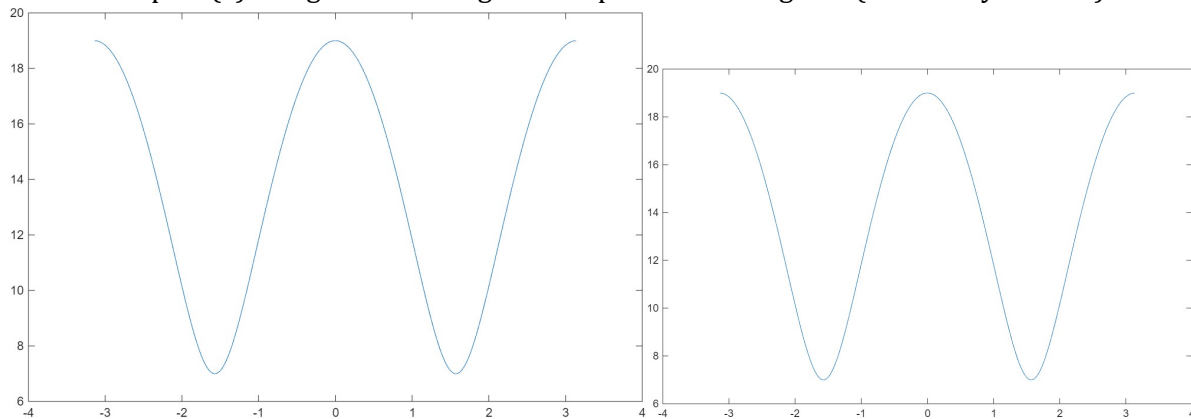
- (a) For $x[n] = 13\delta[n] + 6\delta[n-2]$, use linearity and the time-shift property to find $X(e^{j\omega})$. Identify the phase term(s) due to delay and briefly explain their effect on $|X|$.

$$13 + 6e^{-j\omega 2}$$

- (b) For $x[n] = (-\frac{1}{3})^{n-2} u[n]$, rewrite it to match a known table pair, then derive $X(e^{j\omega})$. State any constraint on the parameter involved.

$$((-1/3)^n * (-1/3)^{-2}) * u[n] \rightarrow (-1/3)^n * 9 * u[n] \rightarrow \frac{9}{1 + \frac{1}{3}e^{-j\omega}}$$

- (c) **(Using MATLAB)** For the signal in part (a), compute a truncated DTFT numerically using $n = -10:10$ and plot $|X(e^{j\omega})|$. On a second figure, plot the magnitude of *your* analytical expression from part (a) using the same ω -grid. Compare the two figures (no overlay needed).



They are very similar and have very little difference, there would be some error as you zoom in as it is an approximation

Question #3: Use the inverse DTFT

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega.$$

- (a) Let $X(e^{j\omega}) = 1$. Evaluate the IDTFT integral and write the resulting $x[n]$. Justify any identity you use.
- (b) Let $X(e^{j\omega}) = C e^{j\omega n_0}$ with $C = 16$ and $n_0 = 23$. Use appropriate properties to find $x[n]$. Briefly explain the time-domain effect of the factor $e^{j\omega n_0}$.
- (c) **(Using MATLAB)** Numerically approximate the IDTFT of part (b) using a uniform grid of ω and a Riemann sum. Plot the reconstructed sequence for $n = -30:30$. From the plot, report the index and amplitude of the dominant sample, and confirm whether it matches *your* analytical result.

Question #4: Relate the continuous-frequency DTFT to the discrete-frequency DFT.

- (a) Explain how the DFT samples the DTFT at $\omega_k = \frac{2\pi k}{N}$. Discuss qualitatively how increasing N changes the frequency resolution and why that helps approximate the DTFT.

$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j(\frac{2\pi}{N})kn} \rightarrow w_k = \frac{2\pi k}{N} \rightarrow$ As N increase, the waveform will more accurately fit the DTFT.

- (b) For $x[n] = (0.9)^{|n|}$, describe qualitatively the behavior of $|X(e^{j\omega})|$ across $\omega \in [-\pi, \pi]$ (e.g., where it is largest/smallest and why). No calculation required—reason from the signal's structure.

There is no phase since it is even / nonzero. Peak at $w = \pi$

- (c) **(Using MATLAB)** Compare DTFT and DFT magnitudes for $x[n] = (0.9)^{|n|}$. Compute a dense DTFT numerically over $\omega \in [-\pi, \pi]$, then plot DFT magnitudes for $N = 32$ on the same axis as discrete markers. In one sentence, state what the markers represent relative to the curve.